# Jonathon **Gardner**

Security Software Engineer

jonathongardner.github.io | jonathongardner | jonathongardner35

## Summary

Security-focused software engineer specializing in binary analysis, credential detection, SBOM automation, and scalable backend systems.

## Skills

**Security and standards:** SPDX, CycloneDX, OCI, TruffleHog, Syft
**Development:** Go, Ruby on Rails, GraphQL, Sidekiq, River
**Data and platforms:** PostgreSQL, MySQL/MariaDB, MongoDB, AWS, Linux, Git
**Binary tooling:** libguestfs, libarchive, go-diskfs, SquashFS

## Work Experience

### Software Engineer
*Feb. 2020 - PRESENT*
Cisco
*Raleigh-Durham, NC*

- Led adoption of Go for high-impact security and systems projects.
- Built a binary credential scanner and integrated it into Cisco release workflows, automating verification and rotation of exposed credentials.
- Re-architected a core firmware extraction tool, reducing processing time for a 5 GB disk image from 24 hours to 2 hours.
- Redesigned the SBOM data model to reduce API latency and support point-in-time tracking.
- Developed an SPDX-compliant Ruby gem to build and sign SBOMs, supporting Cisco's federal software supply-chain requirements.

### Ruby on Rails Software Engineer
*Feb. 2019 - Feb. 2020*
MODIS Contractor for Cisco
*Raleigh-Durham, NC*

- Built Ruby on Rails APIs supporting user and UI workflows.
- Maintained Sinatra and MongoDB microservices.
- Created and maintained Ruby gems to separate concerns across services.

### Ruby on Rails Software Engineer
*Jun. 2017 - Feb. 2019*
ORAU Contractor for Environmental Protection Agency
*Raleigh-Durham, NC*

- Led development of Ruby on Rails APIs for a single-page web application.
- Expanded the Ruby on Rails test suite to identify and eliminate defects.

## Education

### University of South Carolina
*2014 - 2016*
Master of Science in Mechanical Engineering
*Columbia, SC*

- Developed finite element method (FEM) modules in C++ for a MOOSE-based DOE EPSCoR research project.

### Georgia College and State University
*2011 - 2014*
Bachelor of Science in Physics with a minor in Computer Science
*Milledgeville, GA*